

Pedaling Under the Sea: An AI-Assisted Simulation of Electromagnetic Induction

Group 3

Physics II

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Abstract

This project uses AI-assisted design and programming to develop an interactive simulation that demonstrates electromagnetic induction through an animated bicycle-powered generator model. In the simulation, a familiar character pedals a bicycle that provides mechanical input for a generator system. A drive chain connects the bicycle's motion to a separate generator rotor, where north and south magnets rotate inside a pickup coil. As the magnetic field through the coil changes, the simulation displays induced voltage, RMS voltage, current, accumulated energy, magnetic-field behavior, and electron motion in the connected circuit. Users can adjust key generator variables, including pedaling speed, magnet strength, coil area, and number of wire loops. The final version also includes a front bicycle light, an output meter, a jellyfish-style energy indicator, lightning-style feedback when stored energy reaches shock thresholds, and audio controls for background sound and output events. These visual and audio features help users trace the full energy pathway from mechanical input, to changing magnetic flux, to electrical output. Although the simulation is a conceptual educational model rather than a calibrated engineering model, it accurately represents the major cause-and-effect relationships involved in electromagnetic induction. By combining physics modeling, animation, interactivity, and AI-assisted development, the project presents electromagnetic induction in a way that is both engaging and easier for learners to visualize.

Introduction

The physics concept explored in this project is electromagnetic induction. Electromagnetic induction occurs when a changing magnetic flux through a coil produces an induced electromotive force, or voltage. This relationship is described by Faraday's law, which states that the induced voltage depends on how quickly magnetic flux changes through a circuit. In a generator, mechanical motion can rotate a magnet near or inside a coil, causing the magnetic field through the coil to change over time. This changing magnetic field induces voltage and can drive current through a connected circuit. OpenStax describes Faraday's law as the relationship between changing magnetic flux and induced emf, which is the central principle behind this project's generator model (Ling et al., 2016).

Before developing our own simulation, we examined the PhET Generator simulation from PhET Interactive Simulations at the University of Colorado Boulder as shown in Figure 1. The PhET Generator simulation uses moving water to turn a wheel connected to a magnet and coil system, helping users observe how mechanical motion can be converted into electrical output. The simulation is effective because it visually connects mechanical input, magnet rotation, changing magnetic field behavior, and electrical output in one interactive model. It also allows users to adjust variables and immediately observe how those changes affect the system.

Although the PhET Generator simulation clearly demonstrates the science, we wanted to redesign the idea into a more memorable, story-based model. Instead of using a water wheel as the mechanical input, our simulation uses an animated bicycle-powered generator in an underwater scene. Pedaling provides the mechanical input, which turns a drive chain connected to a separate generator rotor. The rotor contains north and south magnets that rotate inside a pickup coil. This design keeps the same core physics idea as the PhET Generator simulation while making the energy pathway easier to follow visually.

The final simulation allows users to adjust pedaling speed, magnet strength, coil area, and number of wire loops. It also includes animated magnetic-field arrows, moving electrons, an optional field meter, an output meter, RMS voltage, instantaneous voltage, current, accumulated energy, a front bicycle light, and a jellyfish-style output indicator. Additional feedback, including lightning effects and sound controls, was added to make changes in electrical output more noticeable. These features were included to help users trace the full pathway from mechanical input, to changing magnetic field, to electrical output.

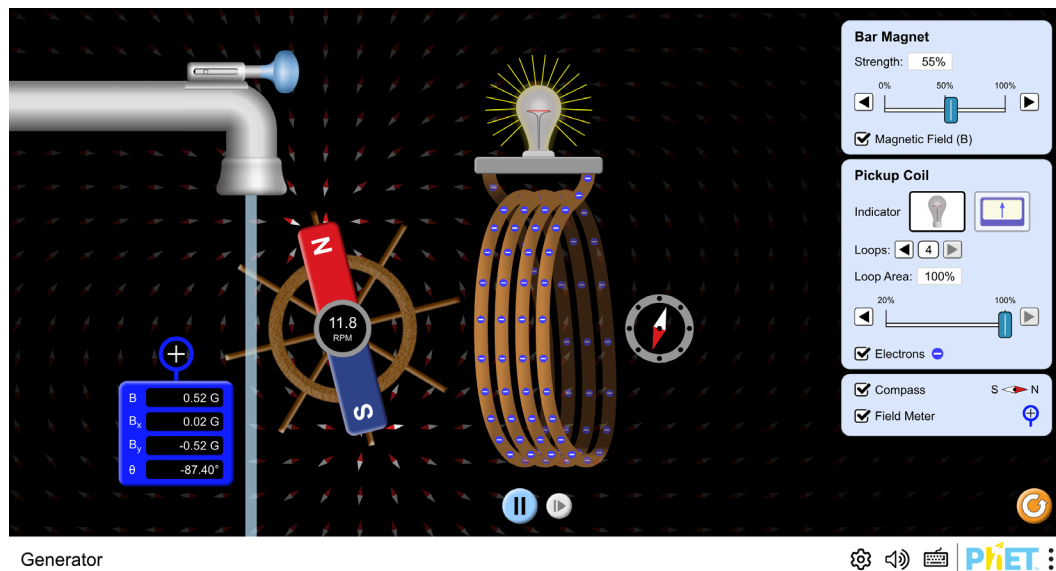


Figure 1. Original PhET Generator simulation showing the water wheel, rotating magnet, coil, magnetic field, and electrical output indicator.

Methods

We developed the simulation through an iterative AI-assisted design process. At the beginning of the project, we tested the same basic task using ChatGPT, Google Gemini, and Claude: recreating the main structure of the PhET Generator simulation. Each AI tool was asked to create an interactive simulation based on the PhET model, including a mechanical input, rotating magnet, coil, magnetic-field visualization, and electrical output. After comparing the results, we chose to continue with the ChatGPT version because it produced the strongest starting point. Its recreation had clearer graphics, smoother animation, more usable controls, and the closest overall resemblance to the original PhET Generator simulation.

After selecting the ChatGPT-generated version, we used it as the foundation for the rest of the project. The first goal was to continue modifying the recreation until it included the same core structure as the PhET Generator simulation. This included showing a mechanical input driving a rotating magnet, a coil responding to the changing magnetic field, field arrows, moving electrons, and output readings. Once the PhET-inspired structure was working as shown in Figure 2, we began redesigning the simulation into an original animated bicycle-powered generator model.

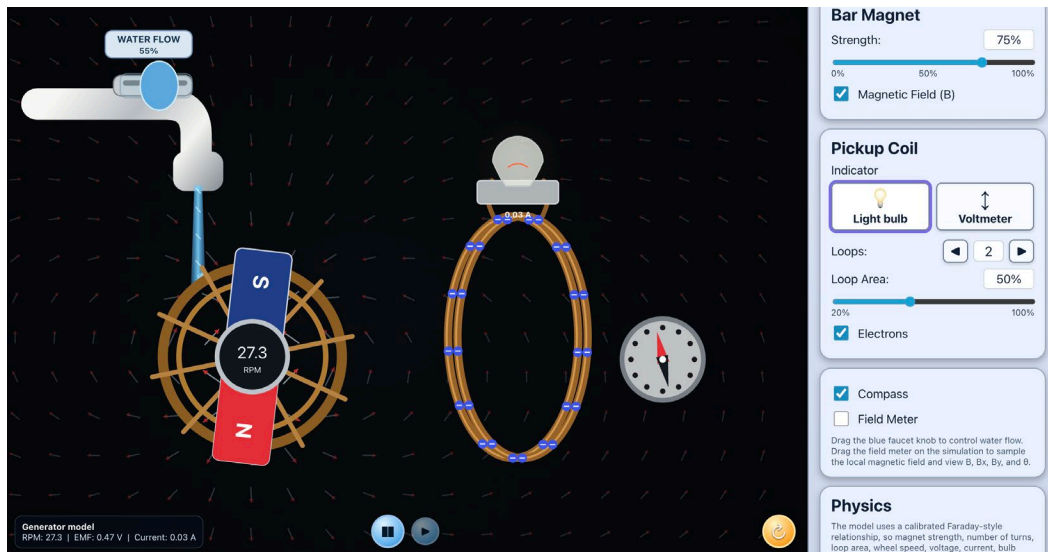


Figure 2. AI-generated recreation of the PhET Generator simulation produced during the first stage of development. This version was used as the foundation for later modifications.

The redesigned version replaced the water-wheel input with a bicycle-pedaling system. In this model, pedaling provides mechanical input, and the chain visually connects the bicycle to the generator system. Several rounds of prompts focused on making the model easier to understand. Early versions placed the magnets directly on the bicycle wheel, but this made the relationship between the magnets and coil less clear. The design was revised, so the bicycle acted as the mechanical energy source while a separate generator rotor with north and south magnets rotated inside the pickup coil. This made the model closer to the conceptual structure of a real generator, where a changing magnetic field through a coil induces voltage and current.

Additional prompts focused on visual clarity and usability. The coil was kept separate from the output indicator, so users could see that the coil is where voltage is induced, while the output machine receives the electrical result. The connecting wires were redesigned, so they looked like part of a continuous circuit connected to the coil and output indicator. Moving electrons were added to the coil and wires to represent the current flow. The final output meter was revised to display instantaneous voltage, RMS voltage, current, and accumulated energy directly on the box, rather than only in a separate heads-up display. The top heads-up display was removed so the main readings would be concentrated on the output meter.

Later, revisions focused on the final user experience. A front bicycle light and jellyfish-style output indicator were added, so electrical output was visible. Lightning effects were added to show when accumulated energy reached certain thresholds, and to enhance engagement. The chain was aligned to the generator rotor, so the mechanical connection was clearer. Audio was also added, including a background loop that plays only while voltage is being generated, a shock sound when the energy threshold effect occurs, and a separate sound when the pedal speed reaches 100 percent. Volume sliders and mute controls were added for each sound. The simulation also stops all audio when the user changes browser tabs, and audio can be enabled by clicking on the simulation.

The simulation was tested by opening each generated HTML version in a web browser and checking whether the controls changed the system in the expected way. Increasing pedaling

speed was expected to increase the rotation rate of the magnetized rotor and increase the electrical output. Increasing magnet strength, coil area, or number of wire loops was expected to increase the generated voltage and current. Screenshots were used throughout the process to identify visual problems, including magnets being off-center, wires not lining up, labels covering parts of the simulation, unclear chain placement, lightning effects that looked too much like a traced outline, audio overlap, and control panels that did not fit the screen well.

AI was helpful for generating code, creating animations, building interactive controls, adding audio features, and revising the simulation across many versions. However, human judgment was still required throughout the project. The AI did not automatically know whether the simulation looked realistic, whether the visuals communicated the physics clearly, whether the audio timing was distracting, or whether the controls were easy to use. Each version had to be reviewed, tested, and corrected. The final simulation was exported as a standalone HTML file titled *Pedaling Under the Sea.html* so it could be opened locally in a browser.

Media Sources and Copyright Considerations

The final simulation includes recognizable visual and audio references inspired by *SpongeBob SquarePants*. These copyrighted media elements are not claimed as original work by the project group. They were used only as part of a noncommercial educational physics demonstration to make the concept of electromagnetic induction more engaging and easier to follow. The purpose of the project is to teach and visualize the physics relationship between mechanical motion, changing magnetic flux, induced voltage, current, and electrical energy output. The *SpongeBob*-related media elements function as decorative and storytelling aids rather than as a substitute for the original television episode or its commercial market.

To reduce copyright concerns, the project does not present the *SpongeBob*-related media as original artwork or original audio, does not sell or commercially distribute the simulation, and uses the material only in the context of an academic demonstration. Because fair use depends on context and is not guaranteed, any public version of the simulation beyond the class or presentation setting should replace these assets with original or properly licensed media, or permission should be obtained from the copyright owner.

Results

The final simulation, as shown in Figure 3 and Figure 4, is an interactive bicycle-powered generator model that demonstrates electromagnetic induction. The simulation begins with an animated character pedaling a bicycle. The bicycle motion acts as the mechanical input to the generator system. A chain connects the pedaling motion to a generator rotor. The rotor contains north and south magnets, and these magnets rotate inside the center of a pickup coil. As the magnetized rotor turns, the magnetic field through the coil changes, inducing voltage and current in the connected circuit.

The simulation includes several adjustable variables. Users can change the pedaling speed, which controls how quickly the rotor turns. Users can also adjust the magnet strength, the number of loops in the pickup coil, and the loop area. These controls allow users to observe how changes in generator design and mechanical input affect the electrical output. The simulation also includes checkboxes for showing or hiding magnetic-field arrows and moving electrons, as well as an

optional field meter for magnetic-field readings. These features make the simulation useful for both simple demonstrations and more detailed explanations.

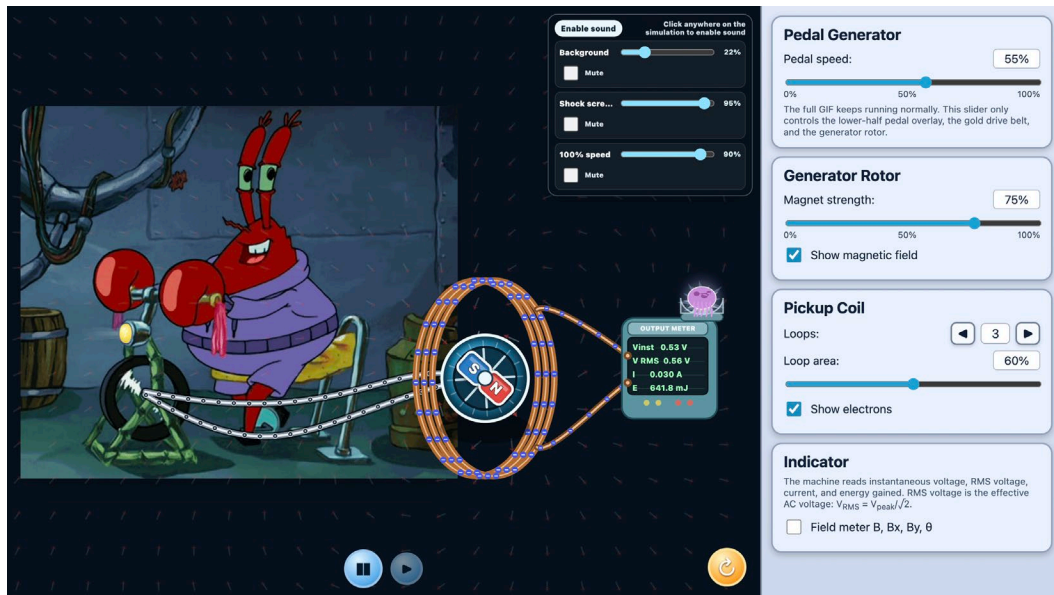


Figure 3. Final animated bicycle-powered generator simulation showing the pedaling input, chain drive, rotating magnetized rotor, pickup coil, magnetic-field arrows, moving electrons, and output indicators.

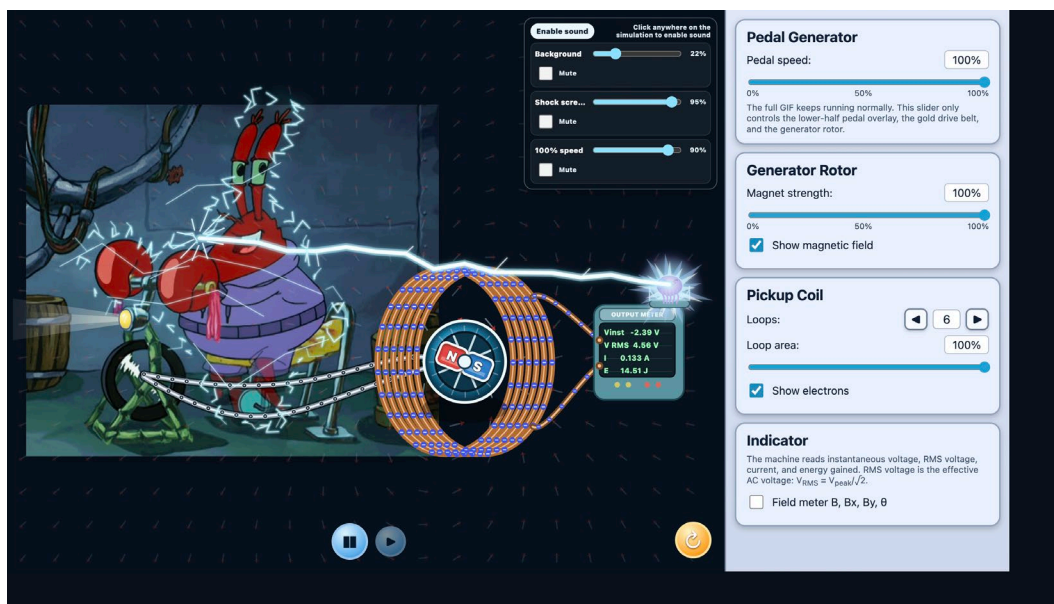


Figure 4. Engaging visual modifications shown in the final simulation

The final design includes multiple visual output indicators. The output meter displays instantaneous voltage, RMS voltage, current, and accumulated energy. The RMS value was included because it represents the effective voltage of the changing AC output, while the instantaneous voltage changes with the rotor's motion. Moving electrons in the coil and wires visually represent current flow through the circuit. Animated magnetic field arrows show how the field changes around the rotating magnetized rotor. A front bicycle light responds to the

generated output, and a jellyfish-style indicator provides an additional visual cue for energy output.

The final version also includes event-based feedback. When the accumulated energy reaches certain thresholds, a lightning effect travels from the output indicator toward the character and creates a surrounding electric effect. This feedback is not meant to be a precise engineering measurement of a real shock. Instead, it is used as a visual representation that the system has generated enough electrical energy to trigger an output event. The thresholds were adjusted so that shocks occur more often at higher energy output, but not so often that the simulation becomes visually or audibly overwhelming.

The audio was added as part of the final interactive experience. A background sound loops while voltage is being generated and stops when the generated voltage is zero. A separate sound plays when the character is shocked, and another sound plays when the pedal speed reaches 100 percent. The final version includes volume sliders and mute checkboxes for the background sound, shock sound, and 100 percent speed sound. The audio also stops when the user changes tabs, preventing sound from continuing when the simulation is not being viewed.

The most important result of the project is that the simulation clearly connects three stages of energy transfer. First, pedaling supplies mechanical input. Second, the mechanical input rotates a magnetized rotor, changing the magnetic field through the coil. Third, the changing magnetic field induces voltage and current, producing electrical output in the connected circuit. This structure makes the simulation more than a simple animation because each visual feature is connected to a physics concept.

Discussion

The main strength of the simulation is that it gives users a visual and interactive way to understand electromagnetic induction. Instead of only presenting Faraday's law as an equation, the simulation shows how each part of a generator contributes to electrical output. Users can see that pedaling alone does not directly create electricity. Instead, pedaling provides mechanical energy; the rotor uses that motion to change the magnetic field through the coil, and the changing magnetic field induces voltage and current in the circuit.

Another strength is the use of adjustable variables. Pedaling speed, magnet strength, coil area, and number of wire loops are all important factors in generator behavior. Allowing users to change these values makes the simulation more interactive and helps users test cause-and-effect relationships. For example, increasing the pedaling speed increases how quickly the magnetic flux changes, which increases the electrical output. Increasing the number of loops or the coil area increases the amount of magnetic flux linked with the coil, which can also increase the induced voltage. Including both instantaneous voltage and RMS voltage also helps show that the generator output changes over time rather than acting like a perfectly steady DC source.

The final simulation is also more engaging because it combines physics modeling with animation, sound, and immediate visual feedback. The moving electrons, magnetic-field arrows, glowing light, output meter, jellyfish-style indicator, lightning effect, and audio controls make the abstract physics easier to follow. These design choices are especially helpful for learners who may struggle to connect the equation for Faraday's law to a physical system.

However, simulation also has limitations. It is a conceptual educational model rather than a fully calibrated engineering model. The voltage, current, RMS voltage, and energy values are simulated to show relationships between variables, not measured values from a real coil, magnet, and load. The model does not include all real-world effects such as friction, heat loss, air resistance, coil resistance changes, battery storage, changing load resistance, or exact generator efficiency. The magnetic-field arrows are visual representations and are not exact field measurements at every point in space.

Another limitation is that the final simulation uses copyrighted SpongeBob-related visual and audio elements to create a familiar theme. These elements helped make the educational model more engaging, but they are not original assets created by the project group. This does not change the physics model, but it does limit how the simulation should be shared. A version intended for public release, wider distribution, or future competition use should replace the copyrighted images and audio with original artwork, original sound effects, or openly licensed media.

Another limitation is that the mechanical drive system is simplified. A real bicycle-powered generator would require more detailed parts such as gears, a belt or chain with tension, bearings, a rotor shaft, and a physical connection that transfers torque. The simulation shows the basic mechanical relationship, but it does not calculate torque, gear ratio, rotational inertia, or mechanical power losses. The shock and lightning effects are also stylized output feedback rather than a literal model of electrical safety or human exposure to electricity.

Future improvements could make the simulation more realistic and more useful for experimentation. A voltage-versus-time graph could be added to show the alternating nature of the induced voltage. A power display could be added using the relationship between voltage, current, and load resistance. The simulation could also include different loads, such as a light bulb, resistor, capacitor, or battery. Adding an efficiency calculation would help users compare mechanical input power to electrical output power. Another improvement would be to allow users to pause the simulation and inspect the magnetic field and current direction at different points in the rotor's motion.

The simulation could also be extended into a lab-style activity. Users could be asked to predict what happens when pedaling speed, magnet strength, coil area, or number of loops is changed. They could then test their predictions using the simulation and record the resulting instantaneous voltage, RMS voltage, current, and energy values. This would make the project stronger as both a demonstration and an educational experiment.

Conclusion

This project successfully developed an AI-assisted interactive simulation of electromagnetic induction using an animated bicycle-powered generator model. The simulation demonstrates how mechanical energy from pedaling can rotate a magnetized rotor inside a pickup coil, creating a changing magnetic field that induces voltage and current in a connected circuit. Users can adjust pedaling speed, magnet strength, coil area, and number of wire loops while observing changes in magnetic-field behavior, electron motion, instantaneous voltage, RMS voltage, current, and accumulated energy.

The final simulation improves the original PhET-inspired concept by using a more engaging visual model and by clearly separating the mechanical input, magnetic-field interaction, and electrical output. The final version also includes a more complete user experience through an output meter, front bicycle light, jellyfish-style energy indicator, lightning feedback, background audio, event sounds, volume controls, and browser tab audio handling. Although the simulation is not a fully calibrated real-world generator model, it accurately represents the main conceptual relationships involved in electromagnetic induction.

Overall, the project demonstrates how AI-assisted programming can be used to create an educational physics simulation that is interactive, visually engaging, and conceptually meaningful. The process also showed that AI tools are most effective when combined with human testing, comparison, revision, and judgment. ChatGPT provided the strongest starting point among the tested AI tools, but the final result required many rounds of human-guided correction and design decisions.

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Media Credits

SpongeBob SquarePants. Created by Stephen Hillenburg, Nickelodeon, 1999-present.